Rooftop Solar Feasibility Study

Deccan Cold Chain Pvt. Ltd.
Pune region warehouses

Prepared by the Projects and Sustainability team for the Board of Directors
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Fictional company and figures, written for training only. Any resemblance to a real company is a coincidence.

1. Executive summary

Deccan Cold Chain runs four temperature-controlled warehouses in the Pune region, at Chakan, Talegaon, Ranjangaon and Bhosari. Together they used about 43 lakh units (kWh) of electricity in the financial year 2025-26, and the electricity bill came to roughly ₹3.96 crore. Electricity is now our second-largest operating cost after staff, and it has grown faster than revenue for three years in a row.

This study looks at whether rooftop solar panels on our own warehouse roofs can bring that cost down, how much it would cost, and how quickly it would pay for itself.

Our main findings:

- The warehouse roofs in the Pune region can host a combined solar capacity of about 1,350 kWp (kilowatt peak) once all roofs are fit for use.
- We recommend a first phase of 1,100 kWp, at an estimated capital cost of ₹4.70 crore, including panels, inverters, mounting, cabling, monitoring and installation.
- Phase 1 is expected to generate about 15.4 lakh units a year, and to save about ₹1.08 crore a year on electricity.
- After yearly maintenance costs, the expected payback period is about 4.6 years. The panels are expected to last at least 25 years.
- The project would avoid about 1,100 tonnes of carbon dioxide emissions a year, which supports the sustainability targets that two of our largest customers now ask us to report on.

We recommend that the Board approves Phase 1, appoints an installation partner through a competitive tender, and starts work in June 2026, so that the first site is producing power before the end of the calendar year.

2. Background and current electricity use

Cold storage runs around the clock. Our compressors, cold rooms, blast freezers and loading-bay doors draw power 24 hours a day, 7 days a week, and our load is highest in the hot months from March to June. Unlike an office, our demand does not fall much at night or on weekends, although weekend loading activity is lower.

Table 1 shows each site's electricity use and bill in the financial year 2025-26.

Site	Opened	Floor area (sq m)	Electricity used (kWh)	Annual bill (₹ crore)
Chakan	2014	18,500	14,50,000	1.33
Talegaon	2017	14,200	11,20,000	1.03
Ranjangaon	2019	12,800	9,80,000	0.90
Bhosari	2009	10,400	7,60,000	0.70
Total		**55,900**	**43,10,000**	**3.96**

The average cost across all four sites works out to about ₹9.20 per unit. Not all of this can be saved by solar. Part of every bill is a fixed charge and a demand charge, which depend on the connected load and the peak demand, not on the units we use. Solar power reduces the energy charge, which for us is about ₹7.00 per unit. All the savings in this report use ₹7.00 per unit, not ₹9.20, so that we do not overstate the benefit.

Over the last three years our electricity cost has grown by about 11 percent a year, while revenue has grown by about 8 percent a year. Two reasons stand out: tariffs have gone up, and we added blast-freezing capacity at Chakan and Talegaon in 2024.

Two of our largest customers, a dairy brand and a frozen-foods company, now ask suppliers to report energy use and emissions each year, and one has said it will prefer suppliers with a clear plan to cut emissions. Rooftop solar is the most visible step we can take on our own.

3. Site assessment

Our engineering team and an independent structural consultant visited all four sites in January and February 2026. At each site they measured the usable roof area, checked the roof's condition and load-bearing strength, looked for shade from nearby buildings, water tanks and equipment, and checked the space available for inverters and cabling.

Chakan

Chakan is our largest and busiest site. The main warehouse has a metal roof installed in 2021, in good condition, with about 4,600 square metres of usable, unshaded area facing south and east. The structure can carry the extra weight of panels and mounting without strengthening. The site has space for inverters next to the existing electrical room. Recommended capacity: 450 kWp.

Talegaon

Talegaon has two roofs. The main roof is in good condition and can take panels. The smaller roof over the dispatch area is shaded for part of the day by a neighbouring building and is not recommended. Usable area is about 3,500 square metres. Recommended capacity: 350 kWp.

Ranjangaon

Ranjangaon is our newest warehouse, and its roof was designed with future solar in mind. The roof is in very good condition and needs no changes. Usable area is about 3,000 square metres. Recommended capacity: 300 kWp.

Bhosari

Bhosari is our oldest site, opened in 2009. The roof sheets on the main building are original, and the structural consultant found corrosion on several purlins and two areas where rainwater has been entering the building. The roof can physically hold about 250 kWp of panels once it is repaired, but it should not carry any additional load in its current state.

For this reason, Bhosari is excluded from Phase 1. The roof repair, estimated at about ₹38 lakh, is already planned in the maintenance budget for the dry season after the 2026 monsoon. Once the repair is complete and signed off by the structural consultant, we will reassess Bhosari for solar as part of Phase 2.

Summary of sites

Site	Usable roof area (sq m)	Roof condition	Recommended capacity (kWp)	Phase
Chakan	4,600	Good	450	1
Talegaon	3,500	Good (main roof only)	350	1
Ranjangaon	3,000	Very good	300	1
Bhosari	2,500	Needs repair	250	2
Total	**13,600**		**1,350**	

4. Technical design

The proposed system at each site has four parts.

Solar panels. We propose high-efficiency monocrystalline panels from a manufacturer with an established service presence in India. Panels would be mounted on a light aluminium frame clamped to the existing metal roof, without drilling through the roof sheets, so that the roof warranty stays valid.

Inverters. Inverters convert the panels' direct current into the alternating current our equipment uses. We propose several string inverters at each site rather than one large central unit, so that a single inverter fault takes only a part of the system offline.

Connection and metering. Each system would connect to the site's main low-tension panel. A bidirectional meter would record both the power we draw from the grid and any surplus solar power we send back to it. This matters because of net metering, which we explain in Section 6.

Monitoring. Each site would have an online monitoring system that shows generation every 15 minutes, raises an alert if output drops unexpectedly, and produces a monthly report for the facilities team.

We expect each kWp of panels to generate about 1,400 units a year in the Pune region. This figure is slightly below typical published values for the region, because warehouse roofs near busy highways collect a lot of dust. The design assumes the panels are cleaned every two weeks, using a waterless cleaning method during the summer months.

For Phase 1, this gives an expected generation of about 15.4 lakh units a year (1,100 kWp × 1,400 units). Output will fall slowly as panels age, by about 0.5 percent a year. The financial figures in this report use first-year generation and do not include this decline, which has only a small effect within the payback period.

Our load is high at all hours, so most solar power will be used on site as it is produced. We estimate that about 69 percent of the solar generation will be used directly by the warehouses. The rest, about 31 percent, is surplus at times when generation is higher than our load, mainly on weekends, on public holidays, and around midday in the cooler months.

5. Costs

Table 3 shows the estimated capital cost of Phase 1. The estimates come from budget quotes by three installers, with a contingency added.

Item	Cost (₹ lakh)
Solar panels	198
Inverters	62
Mounting structures	71
Cabling, protection and metering	48
Monitoring system	9
Installation, testing and commissioning	52
Contingency (about 7 percent)	30
Total, Phase 1 (1,100 kWp)	**470**

This works out to about ₹42,700 per kWp. The figure assumes that the installer is chosen through a competitive tender, and that panel prices stay close to their level in February 2026.

Phase 2 (Bhosari, 250 kWp). We estimate about ₹1.10 crore for the solar system at Bhosari, at today's prices. This does not include the ₹38 lakh roof repair, which is part of the maintenance budget and would be needed whether or not we install solar. Phase 2 is not part of the decision we are asking the Board to make today.

Running costs. Yearly operation and maintenance, including panel cleaning, monitoring, inspections and inverter servicing, is estimated at about ₹6 lakh a year for Phase 1. Inverters usually need replacing once during the life of the panels, typically after 10 to 12 years. We have set aside an estimate of ₹35 lakh for this, in today's money, which is not included in the payback calculation.

Funding. We propose to fund Phase 1 from internal accruals, spread over the financial years 2026-27 and 2027-28. The finance team has confirmed that this does not affect our current loan covenants. We have also looked at a lease model, where an outside company owns the panels and sells us the power at a fixed rate. That option needs no capital from us, but it saves much less over 25 years, and we do not recommend it.

6. Savings and payback

The savings depend heavily on one approval: **net metering.** Under net metering, the surplus power we send to the grid is credited against the units we draw from it at other times. Without net metering, the surplus is either wasted or credited at a very low rate.

Our application for net metering at the three Phase 1 sites has been submitted to the state electricity distribution company. Based on recent applications by other businesses in the same industrial areas, we expect approval by June 2027, but this is not guaranteed.

Table 4 shows the expected yearly savings and payback in two cases.

	With net metering	**Without net metering**
Solar generation (units a year)	15,40,000	15,40,000
Units that reduce our bill	15,40,000	10,62,600 (69 percent)
Value at ₹7.00 per unit (₹ lakh a year)	108	74
Less: operation and maintenance (₹ lakh a year)	6	6
Net yearly saving (₹ lakh)	102	68
Simple payback period	**4.6 years**	**6.9 years**

To be clear: **the 4.6-year payback quoted in the executive summary assumes that net metering is approved by June 2027. If net metering is not approved, the payback period rises to about 6.9 years.** In both cases the project pays for itself well within the life of the panels, but the difference is large enough that the Board should know about it.

Over 25 years, and without assuming any tariff increase, the total net saving with net metering is about ₹23 crore, after allowing for the slow decline in output and one inverter replacement, against an investment of ₹4.70 crore. If electricity tariffs keep rising as they have in recent years, the savings would be higher.

Emissions. Using a grid emission factor of about 0.71 kg of carbon dioxide per unit, Phase 1 would avoid roughly 1,100 tonnes of carbon dioxide a year.

7. Risks

We see five main risks.

1. **Net-metering approval is delayed or refused.** This is the largest financial risk, covered in Section 6. Payback rises from 4.6 years to about 6.9 years. To reduce this risk, our installer will be asked to support the application process, and we will look at shifting some flexible load, such as defrost cycles and battery charging for forklifts, to the middle of the day.
2. **Roof damage or leaks during installation.** A leak in a cold store can damage stock. The tender will require a mounting system that does not drill through the roof sheets, a written roof warranty from the installer, and work only in the dry season.
3. **Panel prices change.** Panel prices have moved sharply in both directions in recent years. The 7 percent contingency covers moderate changes. A larger rise would be reported to the Board before any order is placed.
4. **Lower output than expected.** Dust, shading from new buildings, or poor cleaning can reduce output. Online monitoring and a performance guarantee in the installer contract reduce this risk.
5. **Disruption to operations.** Installation at a working cold store must not block loading bays or cut power to the cold rooms. Work will be planned site by site, with any power shutdowns limited to short, planned windows at night.

8. Implementation timeline

Month	Activity
April 2026	Board approval of Phase 1
May 2026	Tender issued to at least five installers
June 2026	Installer selected, contract signed, detailed design starts
October 2026	Installation starts at Ranjangaon, after the monsoon
November 2026	Ranjangaon commissioned. Installation starts at Chakan
January 2027	Chakan commissioned. Installation starts at Talegaon
March 2027	Talegaon commissioned. Phase 1 complete
June 2027	Expected net-metering approval for all three sites
After roof repair	Bhosari reassessed for Phase 2

Ranjangaon goes first because its roof needs no work and its layout is the simplest, so it is the best place for the installer and our facilities team to learn. Chakan, the largest site, follows once the process has been proven.

9. Recommendation and next steps

The Projects and Sustainability team recommends that the Board:

1. Approves Phase 1, a rooftop solar installation of 1,100 kWp at Chakan, Talegaon and Ranjangaon, with a capital budget of ₹4.70 crore.
2. Authorises the team to run a competitive tender and to select an installer, with the final contract to be signed by the Managing Director and the Chief Financial Officer.
3. Notes that Bhosari is not part of Phase 1, and that a separate Phase 2 proposal will come to the Board after its roof repair.
4. Notes that the expected payback of 4.6 years depends on net-metering approval, and asks the team to report on the approval's progress every quarter.

The team will give the Board a short progress update after the tender, and a full review twelve months after Phase 1 is complete, comparing actual generation and savings with this study.

Appendix A: Key assumptions

Assumption	Value used	Notes
Generation per kWp per year	1,400 units	Slightly below typical values, to allow for dust
Energy charge saved per unit	₹7.00	Fixed and demand charges are not reduced by solar
Share of solar used directly on site	69 percent	Based on half-hourly load data for 2025-26
Operation and maintenance	₹6 lakh a year	Phase 1, all three sites
Panel output decline	0.5 percent a year	Not included in payback figures
Panel life	25 years	Typical manufacturer performance warranty
Grid emission factor	0.71 kg CO_2 per unit	Approximate value used for planning
Expected net-metering approval	June 2027	Not guaranteed, see Section 6

Appendix B: Glossary

kWp (kilowatt peak). The maximum output of a set of solar panels under standard test conditions. A 100 kWp system produces about 100 kW in full, direct sun.

Unit. One kilowatt-hour (kWh) of electricity, the measure used on electricity bills.

Net metering. An arrangement where surplus solar power sent to the grid is credited against power drawn from the grid at other times.

Payback period. The number of years it takes for the yearly savings to add up to the original investment. It does not account for the time value of money.

Inverter. Equipment that converts the direct current from solar panels into the alternating current used by machines and lighting.

Purlin. A horizontal beam in a roof structure that supports the roof sheets.